

Frontend Summer Training

14-Session Course Schedule

HTML · CSS · JavaScript · Bootstrap · React

Incremental E-Commerce Project — ShopFront

Summer 2025 · 42 Hours Total · 14 Sessions × 3 Hours

Phase 1
HTML
Sessions 01–02

Phase 2
CSS
Sessions 03–05

Phase 3
JavaScript
Sessions 06–09

Phase 4
Bootstrap
Sessions 10–11

Phase 5
React
Sessions 12–14

■■ **ShopFront E-Commerce Project** — Built incrementally from a plain HTML page all the way to a fully deployed React application. Every session adds a new feature to the same project.

Phase 1 — HTML

Session 01

HTML Fundamentals & Document Structure

3 hrs

HTML

- How the web works: browsers, servers, HTTP basics
- HTML document structure: DOCTYPE, html, head, body
- Text elements: headings, paragraphs, lists, links
- Semantic HTML5 tags: header, nav, main, section, footer, article

■■ Project Kickoff — Create the skeleton HTML page for ShopFront: header, nav, hero section & footer

Session 02

HTML Forms, Tables & Media

3 hrs

HTML

- Images, audio, video and the figure element
- Hyperlinks: internal, external, anchor links
- Tables: tr, th, td, colspan, rowspan
- Forms: input types, labels, select, textarea, button

■■ Build: Product listing table + Login & Register forms + product image placeholders

Phase 2 — CSS

Session 03

CSS Basics — Selectors, Box Model & Typography

3 hrs

CSS

- CSS syntax, selectors: element, class, id, attribute, pseudo
- The Box Model: margin, border, padding, content
- Typography: font-family, size, weight, line-height, text-align
- Colors: named, hex, rgb, hsl; backgrounds and gradients

■■ Style: Apply base typography, color palette and spacing to the ShopFront page

Session 04

3 hrs

CSS

CSS Layouts — Flexbox & Grid

- Flexbox: container vs items, direction, wrap, justify, align
- CSS Grid: columns, rows, gap, template areas
- Positioning: static, relative, absolute, fixed, sticky
- Overflow, z-index and stacking context

■■ Build: Responsive product grid (3-col desktop → 1-col mobile) using Flexbox & Grid

Session 05

3 hrs

CSS

Responsive Design & CSS Animations

- Media queries and mobile-first strategy
- Viewport units: vw, vh, vmin, vmax
- CSS transitions: property, duration, timing, delay
- CSS animations: @keyframes, animation shorthand
- CSS variables (custom properties)

■■ Polish: Make ShopFront fully responsive; add hover effects on product cards & navbar

Phase 3 — JavaScript

Session 06

3 hrs

JS

JavaScript Fundamentals

- Variables: var, let, const; data types and type coercion
- Operators, conditionals: if/else, switch, ternary
- Loops: for, while, for...of, for...in
- Functions: declarations, expressions, arrow functions, scope

■■ Add: JS product data array + loop to dynamically render product cards to the DOM

Session 07

3 hrs

JS

DOM Manipulation & Events

- Selecting elements: getElementById, querySelector, querySelectorAll
- Manipulating DOM:.textContent, innerHTML, classList, style
- Creating & removing elements dynamically
- Event listeners: click, submit, input, keydown; event object

■■ Build: Add to Cart button updates cart icon count + show/hide cart sidebar

Session 08

3 hrs

JS

Arrays, Objects & Modern ES6+

- Arrays: map, filter, reduce, find, forEach, sort
- Objects: destructuring, spread/rest, optional chaining
- Template literals, short-circuit evaluation
- Modules: import / export (ES Modules)

■■ Build: Product filter by category + search by name using filter() & DOM re-rendering

Session 09

3 hrs

JS

Async JavaScript & APIs

- Callbacks and callback hell
- Promises: .then(), .catch(), .finally()
- async / await syntax
- fetch() API — GET requests, JSON parsing, error handling
- Local Storage: setItem, getItem, removeItem

■■ Fetch real products from FakeStore API; persist cart in localStorage**Phase 4 — Bootstrap**

Session 10

3 hrs **BOOTSTRAP****Bootstrap 5 — Grid, Utilities & Components**

- Bootstrap setup: CDN vs npm; breakpoints overview
- Grid system: container, row, col-* responsive classes
- Utility classes: spacing, colors, display, flex, text
- Components: Navbar, Card, Badge, Alert, Button, Modal

■■ Rebuild: ShopFront layout with Bootstrap grid; Navbar with offcanvas cart + product Cards

Session 11

3 hrs **BOOTSTRAP****Bootstrap Forms, Icons & Customization**

- Bootstrap form controls, validation styles, input groups
- Bootstrap Icons library
- Carousel, Accordion, Tabs, Tooltips & Toasts
- Customizing Bootstrap with CSS variables and Sass overrides

■■ Build: Checkout form with Bootstrap validation + product image Carousel + Toast notifications**Phase 5 — React**

Session 12

3 hrs **REACT****React Fundamentals — Components, JSX & Props**

- Why React? Virtual DOM, component model, SPA concept
- Vite project setup; JSX syntax rules
- Functional components, props, default props
- Rendering lists with .map() and keys; conditional rendering

■■ Migrate: Rebuild ShopFront in React — Navbar, ProductCard & Footer as components

Session 13

3 hrs

REACT

React State, Hooks & React Router

- useState for local state; event handling in React
- useEffect: fetching products from FakeStore API
- Context API — global cart state with useContext
- React Router v6: pages for Home, Products, Product Detail, Cart

■■ Build: Multi-page React shop — live API data, working cart via Context, page navigation

Session 14

3 hrs

REACT

Final Polish, Deployment & Project Showcase

- React Hook Form — checkout form with validation
- Performance basics: React.memo, lazy loading pages
- Production build with Vite; deploying to Vercel/Netlify
- Project retrospective: what we learned & next steps

■■ FINAL: Deploy ShopFront live to Vercel — full working e-commerce app demo & showcase!

Course At a Glance

#	Title	Phase	Tech	Hrs
01	HTML Fundamentals & Document Structure	HTML	HTML	3
02	HTML Forms, Tables & Media	HTML	HTML	3
03	CSS Basics — Selectors, Box Model & Typography	CSS	CSS	3
04	CSS Layouts — Flexbox & Grid	CSS	CSS	3
05	Responsive Design & CSS Animations	CSS	CSS	3
06	JavaScript Fundamentals	JavaScript	JS	3
07	DOM Manipulation & Events	JavaScript	JS	3
08	Arrays, Objects & Modern ES6+	JavaScript	JS	3
09	Async JavaScript & APIs	JavaScript	JS	3
10	Bootstrap 5 — Grid, Utilities & Components	Bootstrap	BOOTSTRAP	3
11	Bootstrap Forms, Icons & Customization	Bootstrap	BOOTSTRAP	3
12	React Fundamentals — Components, JSX & Props	React	REACT	3
13	React State, Hooks & React Router	React	REACT	3
14	Final Polish, Deployment & Project Showcase	React	REACT	3

■ **Tech Stack & Tools:** HTML5 · CSS3 · JavaScript ES6+ · Bootstrap 5 · Bootstrap Icons · React 18 · Vite · React Router v6 · React Hook Form · Context API · FakeStore API · localStorage · Vercel